ROCULTURE AI

- Digital Encyclopedia of Romania -

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Introduction

The ROCULTURE AI project aims to develop an intelligent platform for promoting and valorizing Romanian cultural heritage using advanced AI technologies. The platform focuses on digitizing, classifying, and contextualizing cultural resources - both tangible and intangible - making interactive information accessible to the public

The platform aspires to be a modern reference for preserving and transmitting cultural values, ensuring open and sustainable access to heritage in an innovative digital ecosystem.

Technologies

Core technologies of the project:

- **Backend:** Written in Python and served via FastAPI.
- **Frontend:** Built with plain HTML and JavaScript.
- **Database:** ChromaDB, used for storing and querying vector embeddings.
- **Hosting:** Cloudflare routes domain to local server.
- **Image Processing:** Images are converted to embeddings using clip-vit-base-patch32.
- **Text Processing / Chatbot:** *Gpt-5-nano* and *gpt-4* for analyzing text, generating responses and *text-embedding-3-large* creating text embeddings.

Embeddings are numerical vectors that represent the meaning of images or text. Instead of storing raw data, the system converts content into these vectors so it can compare, search, and classify information efficiently.

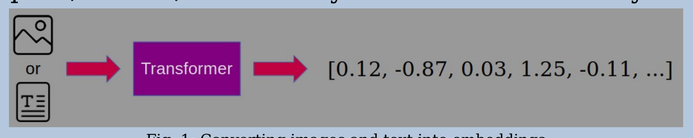


Fig. 1: Converting images and text into embeddings

Hosting Approach

The site is currently hosted on a local machine as a prototype, which serves the website. The domain is connected to Cloudflare, securely forwarding visitors' requests to the local server. A Cloudflare Tunnel ensures a protected connection, allowing users to access the live site from anywhere.

The site is hosted at uwudoll.com.

AI-Driven Content Generation

The site's content is managed by a administrator, who adds new objects by creating folders on the server and uploading relevant images. Images are added manually to ensure accuracy and maintain quality control, as automated image collection could lead to inconsistent or irrelevant content.

Once images are added, a script generates embeddings for each image, converting them into numerical representations suitable for similarity search.

In addition, another script gathers information about each object from the web and generates descriptive text. The resulting description is then divided into smaller text chunks, which are converted into embeddings using text-embedding-3-large and stored locally in the database. This process ensures that both images and textual content are searchable and retrievable efficiently.

Data Flow and Processing Diagram

This diagram illustrates how data moves through the system:

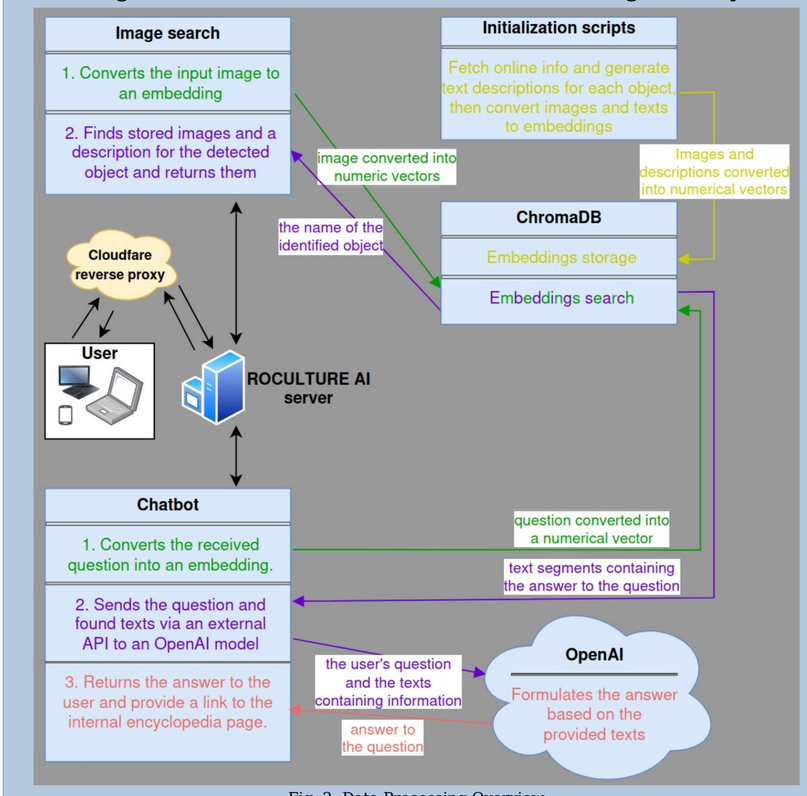


Fig. 2: Data Processing Overview

Site description

The site consists of three main pages, each offering a distinct way to explore Romanian cultural heritage

Image search: Users can upload an image, which is processed locally into an embedding. This embedding is then used to query ChromaDB to find the most similar object related to Romania,The system returns both images and textual information about the identified object, allowing for efficient retrieval of culturally relevant content.

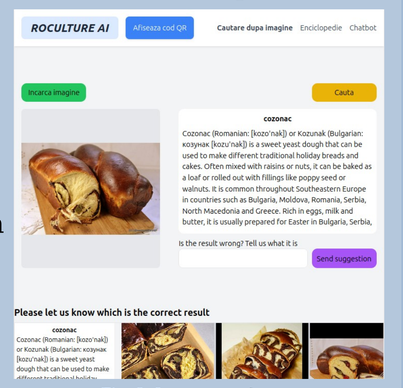


Fig. 3: Image search page

Encyclopedia: This page allows users to freely browse all objects stored in the database, viewing associated images and textual descriptions. Objects are organized into three categories - Gastronomy, Locations, and Traditions - providing structured access to Romania's cultural resources.

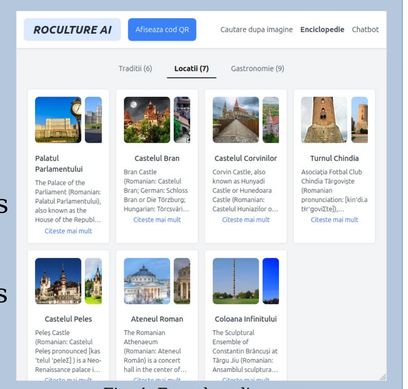


Fig. 4: Encyclopedia page

Chatbot: Users can type questions about Romania. The system uses embeddings to search the local database for texts likely containing the answer. These texts are then sent to the gpt-5-nano API, which generates a summarized response that is presented to the user, combining AI-driven analysis with curated cultural data.

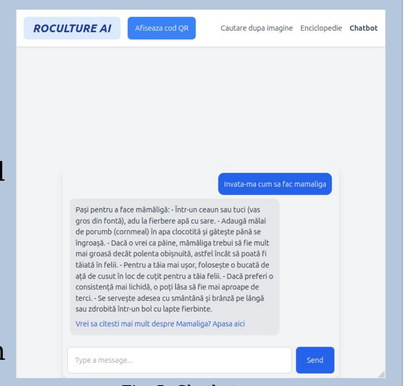


Fig. 5: Chatbot page

Users can navigate between pages by clicking the links in the header. Additionally, each page features a QR code that, when scanned, redirects directly to that specific page for convenient access.